

ONE-SAMPLE DISTRIBUTION-FREE INFERENCE ON DISPERSION

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SUMMARY

The interquantile distance covering the central 90 per cent of the probability mass of a distribution is regarded as a general measure of dispersion. Chu (1957) proposed distribution-free inferential methods for such interquantile distances for the one-sample problem. A procedure closely related to that of Chu, but substantially more powerful, is developed.

Key words: Dispersion, distribution-free, one-sample problem.

1. Introduction

Let $Y_1 < \dots < Y_n$ denote the order statistics of a random sample from the continuous distribution with c.d.f. $F(\cdot)$. The p -th quantile of $F(\cdot)$ is denoted by ξ_p .

The length $\xi_{1-p} - \xi_p$, $0 < p < \frac{1}{2}$, of the central interval containing $(1-2p)$ of the probability mass is the general parameter of interest.

Chu (1957) deduced confidence intervals for $\xi_{1-p} - \xi_p$ of the form $(Y_{n+1-r} - Y_r, Y_{n+1-s} - Y_s)$, $r > s$, which also provide a means of testing hypotheses regarding $\lambda_p \equiv \xi_{1-p} - \xi_p$. Chu paid no attention to the value of p .

The problem of finding the best r for setting one-sided confidence intervals for the standard deviation σ of F , of the form $\alpha(Y_{n+1-r} - Y_r)$, was treated by Leone, Rutenber and Topp (1961). Specifically for uniform, normal and exponential distributions $\frac{r}{n}$ is generally small.

As to be expected the best point estimators of σ of the form $\alpha(Y_{n+1-r} - Y_r)$ are also obtained when $\frac{r}{n}$ is small. Specifically for large n the optimum values of $\frac{r}{n}$, for uniform, normal and exponential distributions, range between as small as possible and 0.13. Similar results hold for small sample sizes. (Section 3).

Instead of relating $\xi_{1-p} - \xi_p$ to the standard deviation, it can be regarded as a measure of spread in its own right, and contrary to the standard deviation $\xi_{1-p} - \xi_p$ has the same probabilistic interpretation for all continuous distributions. Broadly speaking $\xi_{1-p} - \xi_p$ is a distribution-free measure of dispersion and this makes it particularly suited to distribution-free inference. Heuristically, for $\xi_{1-p} - \xi_p$ to be a meaningful measure of dispersion, p should not be too small for then $\xi_{1-p} - \xi_p$ will not effectively

discriminate between distributions with infinite range. On the other hand if p is large (say $p=0.3$) then sixty percent of the probability mass, in a certain sense, is *not* controlled by this measure. In the light of these heuristic considerations, together with the theoretical indications above, $p=0.05$ is proposed and $\lambda = \xi_{0.95} - \xi_{0.05}$ formally defined as the measure of dispersion.

2. Test Statistics

Consider the testing problem where H_0 specifies $\xi_{0.05}$ and $\xi_{0.95}$ as $\xi_{0.05}^0$ and $\xi_{0.95}^0$ while the alternative is $H_1: \lambda > \xi_{0.95}^0 - \xi_{0.05}^0$. Let T denote the number of observations below $\xi_{0.05}^0$ or above $\xi_{0.95}^0$ then $\{t: t \geq c\}$ with c determined by

$$B_n(c-1; 0, 10) \geq 1 - \alpha > B_n(c-2; 0, 10)$$

is a logical α -level critical region for testing H_0 against H_1 .

$$(B_n(c; p) = \sum_{r=0}^c \binom{n}{r} p^r (1-p)^{n-r}).$$

In general the null hypothesis will be composite in the sense that only the value of λ will be specified, say $H_0: \lambda = \lambda_0$. In this case estimate the position of the interval with length λ_0 such that as many observations fall above it as below it and let T^* denote the number of observations outside this interval. Three test procedures are available.

For any continuous distribution the size of the critical region $C_1 = \{t^*: t^* \geq 2c_1\}$ for testing H_0 , with c_1 determined by

$$B_n(c_1 - 1; 0, 05) \geq 1 - \frac{\alpha}{2} > B_n(c_1 - 2; 0, 05),$$

is at most α . An equivalent form of C_1 is

$$C_1 = \{y_{n+1-c_1} - y_{c_1} : y_{n+1-c_1} - y_{c_1} > \lambda_0\}.$$

For $f(x) = F'(x)$ possessing a derivative continuous in some neighbourhood of $\xi_{0.05}$ and $\xi_{0.95}$ the size of the critical region $C_2 = \{t^*: t^* \geq 2c_2\}$ for testing H_0 , with c_2 determined by

$$B_n(c_2 - 1; 0, 05) \geq 1 - \alpha > B_n(c_2 - 2; 0, 05),$$

is asymptotically at most α . An equivalent form of C_2 is

$$C_2 = \{y_{n+1-c_2} - y_{c_2} : y_{n+1-c_2} - y_{c_2} > \lambda_0\}.$$

For $f(x)$ symmetric and possessing a derivative continuous in some neighbourhood of $\xi_{0,05}$ and $\xi_{0,95}$ the size of the critical region $C_3 = \{t^* : t^* \geq c_3\}$ for testing H_0 , with c_3 determined by

$$B_n(c_3 - 1; 0, 1) \geq 1 - \alpha > B_n(c_3 - 2; 0, 1),$$

is asymptotically α .

Inspection shows that for $\alpha \leq 0,5$ $C_1 \subset C_2 \subset C_3$ and hence, when applicable, C_3 will be uniformly better than C_2 and C_2 uniformly better than C_1 . Due to the restriction of symmetry C_3 will not be of much practical use, especially since the size of C_3 can be substantially in excess of the nominal α for extreme skew distributions. The procedure based on C_1 , which is basically that of Chu (1957), is compared empirically with that based on C_2 for uniform, normal and exponential distributions. Both procedures are conservative while in terms of power, against $H_1 : \lambda > \lambda_0$, C_2 is substantially better than C_1 . (Section 5).

On rejecting H_0 a lower confidence bound for λ is provided by $y_{n+1-c_2} - y_{c_2}$ with an asymptotic confidence level of at least $1 - \alpha$.

Critical regions for the alternative hypotheses $\lambda < \lambda_0$, $\lambda \neq \lambda_0$ can similarly be constructed. Since $P(T^* = 0 | H_0)$ will exceed α for small n one must either use randomization or only apply the test for large n . Due to $\lambda > \lambda_0$ being by far the most important alternative in practice, this disadvantage is not too serious.

3. Point Estimation of σ based on Interquantile Distances

Consider the unbiased estimator of σ

$$c_{n,r}(Y_{n+1-r} - Y_r)$$

with

$$c_{n,r} = \sigma / E(Y_{n+1-r} - Y_r).$$

In terms of minimum variance the best estimator of this form is that for which

$$D^2 = \text{var}(Y_{n+1-r} - Y_r) / \{E(Y_{n+1-r} - Y_r)\}^2$$

is minimized with respect to r .

From the moments of order statistics, Sarhan & Greenberg (1962), the results in table I show that r/n should be small.

Furthermore for $n \rightarrow \infty$ and r such that $0,5 > \lim_{n \rightarrow \infty} \frac{r}{n} = p > 0$ the asymptotic distribution of $Y_{n+1-r} - Y_r$ is normal with mean $\xi_{1-p} - \xi_p$ and variance

$$p(1-p) \{ (1/f^2(\xi_{1-p}) + 1/f^2(\xi_p)) \} / n - 2p^2 / \{ f(\xi_{1-p})f(\xi_p)n \}$$

and the values of p minimizing D^2 for uniform, normal and exponential distributions are readily found to be: As small as possible, 0,07 and 0,13 respectively.

TABLE I

Value of D^2 for samples of size n from different distributions

n	r	Uniform	Normal	Exponential
5	1	0,0714	0,1380	0,3280
	2	0,2857	0,3297	0,5200
10	1	0,0185	0,0671	0,1924
	2	0,0476	0,0853	0,1787
	3	0,1000	0,1301	0,2192
20	1	0,0048	0,0381	0,1266
	2	0,0107	0,0366	0,0949
	3	0,0182	0,0406	0,0898
	4	0,0280	0,0478	0,0932
	5	0,0413	0,0589	0,1028

4. Verification of Inferential Statements

Firstly

$$P(Y_{n+1-c_1} - Y_{c_1} > \lambda_0 \mid H_0) = P(Y_{n+1-c_1} - Y_{c_1} > \xi_{0,95}^0 - \xi_{0,05}^0)$$

and from Chu (1957) this is less than or equal to $2(1 - B_n(c_1 - 1; 0,05))$. The procedure based on C_1 is thus basically that of Chu.

Secondly consider

$$P(T^* \geq 2np + d\sqrt{n})$$

with $p = 0,05$ and d an arbitrary constant. Under H_0 this can be written as

$$P(Y_{n+1-r_n} - Y_{r_n} \geq \xi_{1-p} - \xi_p)$$

where r_n is the smallest integer such that $r_n \geq (2np + d\sqrt{n})/2$.

If $f(x) = F'(x)$ possesses a derivative continuous in some neighbourhood of ξ_{1-p} and ξ_p then the asymptotic distribution of $Y_{n+1-r_n} - Y_{r_n}$ is normal with mean $\xi_{1-p_n} - \xi_{p_n}$ where $p_n = r_n/n$, and variance

$$p_n(1-p_n) \{1/f^2(\xi_{p_n}) + 1/f^2(\xi_{1-p_n})\}/n - 2p_n^2 \{nf(\xi_{p_n})f(\xi_{1-p_n})\}.$$

These expressions simplify to

$$\xi_{1-p} - \xi_p - d \{1/f(\xi_p) + 1/f(\xi_{1-p})\}/2\sqrt{n} + o(1/\sqrt{n})$$

and

$$p(1-p) \{1/f^2(\xi_p) + 1/f^2(\xi_{1-p})\}/n - 2p^2 \{nf(\xi_p)f(\xi_{1-p})\} + o(1/n).$$

Hence, for all d ,

$$\lim_{n \rightarrow \infty} P(T^* \geq 2np + d\sqrt{n}) = \lim_{n \rightarrow \infty} P(Z \geq d\sqrt{n}/V_n)$$

with

$$V_n^2 = 4n[p(1-p) \{1/f^2(\xi_p) + 1/f^2(\xi_{1-p})\} - 2p^2 \{f(\xi_p)f(\xi_{1-p})\}].$$

$$[1/f(\xi_p) + 1/f(\xi_{1-p})]^{-2}$$

which proves the asymptotic distribution of T^* to be normal with mean $2np$ and variance V_n^2 . A more convenient form for V_n^2 is

$$V_n^2 = n(2p)(1-2p) + n(2p) \{1/f(\xi_p) - 1/f(\xi_{1-p})\}^2 / \{1/f(\xi_p) + 1/f(\xi_{1-p})\}^2.$$

For symmetrical distributions V_n^2 reduces to $n(2p)(1-2p)$ in which case T^* and T have the same asymptotic distribution under H_0 and this gives rise to C_3 . Finally the upper bound of V_n^2 is

$$n(2p)(1-2p) + n(2p) = 4np(1-p)$$

and hence in the extreme case of this upper bound $\frac{T^*}{2}$ and the binomial variate W with parameters n and $p=0.05$ asymptotically have the same distribution. For $c > np$ it thus follows that

$$\lim_{n \rightarrow \infty} P(T^* \geq 2c | H_0) \leq \lim_{n \rightarrow \infty} P(W \geq c)$$

which establishes C_2 .

5. Empirical Comparison of C_1 , C_2 and C_3

The critical regions C_1 and C_2 are compared empirically (1 000 repetitions) for sample sizes 5, 10, 20 from uniform, normal and exponential distributions. Throughout both critical regions are conservative, C_2 being substantially more powerful than C_1 . For the uniform and normal distributions the critical region C_3 , which requires the assumption of symmetry, is similarly more powerful than C_2 .

The results in Table 2 are illustrative.

TABLE 2

Empirical power of C_1 , C_2 and C_3 for sample size n from different distributions;

$$\alpha = 0,05^*; H_0: \xi_{0,05} = \xi_{0,05}^0, \xi_{0,95} = \xi_{0,95}^0 \text{ and } \pi = F(\xi_{0,05}^0) + 1 - F(\xi_{0,95}^0).$$

π	n	Uniform			Normal			Exponential	
		C_1	C_2	C_3	C_1	C_2	C_3	C_1	C_2
0,10	5	0,001	0,011	0,047	0,003	0,021	0,087	0,013	0,032
0,20	5	0,015	0,049	0,169	0,017	0,063	0,228	0,036	0,074
0,30	5	0,033	0,088	0,281	0,049	0,199	0,368	0,111	0,171
0,50	5	0,203	0,278	0,546	0,243	0,321	0,596	0,376	0,438
0,10	10	0,003	0,007	0,014	0,005	0,031	0,026	0,014	0,033
0,20	10	0,026	0,068	0,129	0,036	0,087	0,160	0,055	0,104
0,30	10	0,093	0,195	0,343	0,129	0,245	0,415	0,222	0,343
0,50	10	0,485	0,630	0,840	0,493	0,647	0,869	0,647	0,750
0,10	20	0,002	0,004	0,015	0,002	0,007	0,024	0,015	0,033
0,20	20	0,059	0,134	0,237	0,071	0,163	0,287	0,089	0,174
0,30	20	0,263	0,405	0,574	0,333	0,486	0,662	0,465	0,585
0,50	20	0,876	0,926	0,979	0,915	0,951	0,989	0,949	0,969

* Randomization has been applied to make the sizes of the tests exactly α .

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